Q.24 Define coplanar and non coplanar force system.

Q.25 State triangle law of forces.

Q.26 Define action and reaction.

Q.27 Explain strain and types of strain.

Q.28 State polygon law of forces.

Q.29 Describe load and effects of a load on a member.

Q.30 Explain Hook's Law.

Q.31 Define parallel Axis theorem.

Q.32 Differentiate between centroid and centre of gravity(Any five)

Q.33 Define maximum and premissible bending stress.

Q.34 Define point of contraflecure.

Q.35 Define uniformly varying load and uniformly distributed Load.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 A cantilevel beam of span 6m carries a uniformly distribute load of 3KN per m over the whole span. Draw its shear force and bending moment diagram.

Q.37 Draw a typical stress strain curve for a mild steel specimen subjected to a tensile force and mention briefly its salient features.

Q.38 Explain Lami's Theorem and Varignon's Theorem.

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4th Sem / Arch
Subject:- Structural Mechanics

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 A body which does not change its shape and size under effect of forces is called

a) rigid body b) soft body

c) hard body d) very hard body

Q.2 Moment is a_____ quantity.

a) Scalar b) vector

c) static d) dynamic

Q.3 Force is measured by product of

a) mass & velocity

b) mass & acceleration

c) weight & acceleration

d) momentum & velocity

Q.4 Which of the following is a vector quantity?

a) mass b) volume

c) density d) none of the above

Q.5 Every material obey's the Hook's law within

a) elastic limit

b) plastic limit

c) limit of proportionality

d) none of these

Q.6 The property of a material by which it can be beaten or rolled into sheets is called

a) elasticity b) plasticity

c) ductility d) malleability

Q.7 Which of the following material is more elastic?

a) rubber b) glass

c) steel d) wood

Q.8 If a material has identical properties in all directions, it is called

a) elastic b) plastic

c) Isotropic d) homogeneuous

Q.9 At the point of contraflexure

a) B.M. is maximum b) B.M. is minimum

c) B.M. is zero d) none of these

Q.10 Brittleness is opposite to

a) Toughness b) Proportional limit

c) plastic limit d) yield point

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 In SI system, the unit of weight is_________.

Q.12 A force can be resolved into_______ number of pairs of components.

Q.13 The forces which act at a single point is known as_________.

Q.14 Resultant of two forces is minimum when the angle between them is________.

Q.15 The unit of strain is_______.

Q.16 Shear strain is expressed in_______.

Q.17 The negative bending moment is called______ moment.

Q.18 Point of contraflexure is also known as point of ________.

Q.19 A continuous beam is supported on more than_____ supports

Q.20 The unit of radius of gyration is_______.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Describe pin jointed frames.

Q.22 Differentiate between point load and U.D.L (any five with diagram)

Q.23 Define Lami's Theorem.